LOCAL STRUCTURE OF ÉTALE ALGEBRAS: THE PROOF AS FORMALISED IN UNIT

Abstract. Let R be a residually discrete local ring, E an étale R-algebra and $\mathfrak{q}$ a decidable prime of E above $\mathfrak{m}$. Then $E[1/s']$ is standard étale for some $s' \notin \mathfrak{q}$. This is the local theorem of the note *Local structure of étale algebras* (`zmt2-revised.tex`). We give its proof in the form that has been checked in Unit. That form simplifies the note at several points: a remainder loop replaces gcds, there is no gcd with X, Section 1 is read in E itself, and the derivative step uses dual numbers. The last section relates the proof to the formal development, where Zariski's Main Theorem is no longer a hypothesis: it is the theorem proved in `sag/zmt-general`. The required field theorem is now checked too: a finitely presented, formally unramified algebra over a discrete nonzero field has a separable monogenic neighbourhood of a specified decidable prime. The full finite product of separable monogenic algebras is also checked, with no prime input. The local adapter constructs its basis and selected factor using the selected theorem. Connecting all the adapter's explicit local inputs to the original hypotheses remains unfinished. The same local conclusion is checked for E finitely presented, flat and unramified, subject to those inputs, with an ending that replaces the special-fibre comparison and Nakayama's lemma of the note by a flatness transfer theorem. Separately, `sag/etale-direct` now proves over an arbitrary base that finite presentation, flatness and formal unramifiedness imply unique square-zero lifting, without ZMT or a covering argument. Its multivariable Jacobian proof does not supply the one-variable standard charts of the local structure theorem; see `etale-direct-analysis.tex`.

1. Definitions and statement

All rings are commutative. An R-algebra E is *étale* if it is finitely presented and *formally étale*. That is, for every R-algebra C and square-zero ideal $N \subseteq C$, every algebra map $E \to C/N$ lifts uniquely to $E \to C$. We separate the two halves: *lifting* (existence) and *unramified* (uniqueness). Flatness is not part of the definition.

For $Q, G \in R[X]$ with Q monic, a *point* of (Q, G) in an R-algebra C is an element $a \in C$ with $Q(a) = 0$ and $G(a)Q'(a)$ a unit. The algebra

$$S = (R[X]/(Q))[1/(GQ')(a)], \qquad a = X \bmod Q,$$

is characterised by its points: S has a point a, and $h \mapsto h(a)$ is a bijection from algebra maps $S \to C$ to points of (Q, G) in C. An algebra is *standard étale* if it is isomorphic to such an S. A standard étale algebra is unramified: two maps that agree modulo a square-zero ideal send a to roots $a_1, a_2 = a_1 + \varepsilon$ with $\varepsilon^2 = 0$. Taylor's formula gives $0 = Q(a_2) - Q(a_1) = Q'(a_1)\varepsilon$, and $Q'(a_1)$ is a unit, so $\varepsilon = 0$.

R is *residually discrete local* with maximal ideal $\mathfrak{m}$: every $a \in R$ is in $\mathfrak{m}$ or a unit, and $1 - a$ is a unit for $a \in \mathfrak{m}$.

Theorem 1.1 (Local structure)**.** *Let E be an étale R-algebra and $\mathfrak{q}$ a prime of E containing $\mathfrak{m}E$, with decidable membership. There is $s' \in E \setminus \mathfrak{q}$ such that $E[1/s']$ is standard étale over R.*

The proof has two parts. Sections 2–6 build a monic Q and a point y of (Q, G) in a localisation of E. This part uses only that E is finitely presented and that its special fibre is finite étale. Section 7 then uses that E is étale. Section 8 replaces this last part when E is flat and unramified instead.

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2. The two inputs

The proof uses ZMT and a field theorem. Both have now been proved in Unit in the forms needed here. Their final local-base integration is described in Section 10.3; the original local interfaces below still take the corresponding data explicitly.

Zariski's Main Theorem [1, 3]. Let $x_1, \dots, x_n$ generate E, and let $E/\mathfrak{m}E$ be finite over $k = R/\mathfrak{m}$. Then there is $s \in 1+\mathfrak{m}E$ such that $B = R[s, sx_1, \dots, sx_n] \subseteq E$ is a finite R-module, with generators $b_1, \dots, b_r$; and $B[1/s] = E[1/s]$. Only the following consequences are used: $s \notin \mathfrak{q}$, because $\mathfrak{m}E \subseteq \mathfrak{q}$ and $1 \notin \mathfrak{q}$; and every $z \in E$ satisfies $s^k z \in B$ for some k, since z is a polynomial in the x_i. The proof does not use that $B \to E$ is injective, so we write $j : B \to E$ only where the distinction matters. The coordinates of an element of B in the b_i, and the k with $s^k z \in B$, are known to exist but are not given by a function. At each stage the proof uses finitely many of them, and its conclusion is a proposition, so this is enough (Section 10).

The original field interface [5, 4] starts with a finite product of monogenic finite étale factors $k[X]/(P_i)$ of $(E/\mathfrak{m}E)[1/s]$, $k = R/\mathfrak{m}$. Only the following consequences are used. There are $\varepsilon_i, x_i \in B$ and $P_i \in R[X]$ such that, for some N:

- $s^N(\sum_i \varepsilon_i - 1) \in \mathfrak{m}E$;
- $s^N \varepsilon_i P_i(x_i) \in \mathfrak{m}E$ (root);
- $s^N \varepsilon_i (P_i'(x_i) w_i - 1) \in \mathfrak{m}E$ for some w_i (separable);
- for every $b \in B$ there is $C \in R[X]$ with $s^N \varepsilon_i (b - C(x_i)) \in \mathfrak{m}B$ (generation).

Orthogonality and idempotency of the ε_i are not used, nor is monicity of P_i.

The checked selected-factor field theorem avoids constructing this product. It supplies a single factor outside the specified prime, and polynomial representations only for a finite basis. The refined local proof needs only the representations of the finitely many b_j. Thus the choice of a factor in the next lemma can be performed within the field theorem itself. The coefficient and denominator bridge is checked, with the remaining explicit local inputs listed in Section 10.3. The original full product is now independently constructed in `sag/etale-field-product`, without a specified prime. Its factors are actual monic quotients with invertible derivative, and the product comes with inverse algebra maps. Its integration into the original factor-selection route remains separate from the existing selected local adapter.

Lemma 2.1 (Choice of a factor)**.** *Some ε_i is outside $\mathfrak{q}$. With $s' = s\varepsilon_i$, all the data above hold for s' and the single factor (x_i, P_i).*

Proof. Decide $\varepsilon_i \in \mathfrak{q}$ for each i. If all are in $\mathfrak{q}$, so is $\sum \varepsilon_i$, hence so is s^N, since $s^N(\sum \varepsilon_i - 1) \in \mathfrak{m}E \subseteq \mathfrak{q}$. But $s \notin \mathfrak{q}$ and $\mathfrak{q}$ is prime. For the refinement use $(s\varepsilon)^{N+1} w = (s\varepsilon^N)\big(s^N(\varepsilon w)\big)$ and $(s\varepsilon)^k z = \varepsilon^k (s^k z)$. □

From now on there is one factor: $s \notin \mathfrak{q}$, $x \in B$ and P with

$$s^N P(x) \in \mathfrak{m}E, \qquad s^N(P'(x)w - 1) \in \mathfrak{m}E, \qquad \forall b\ \exists C :\ s^N(b - C(x)) \in \mathfrak{m}B. \tag{2.1}$$

We may assume $x \notin \mathfrak{q}$: otherwise $x - 1 \notin \mathfrak{q}$, and we replace x by $x - 1$ and P by $P(X + 1)$.

3. An idempotent and the determinant lemma

Fix an ideal I of B containing $\mathfrak{m}B$.

Lemma 3.1 (Residual idempotent)**.** *Suppose $\sum_i c_i t^i \in I$ for some $t \in B$, where some c_i is a unit. Then there are v and $N \geq 1$ with $t^N(1 - tv) \in I$. Put $e = (tv)^N$. Then $e^2 \equiv e$, $t^N(1 - e) \in I$, and $e \in tB$. Moreover $t^K w \in I$ implies $ew \in I$.*

Proof. Using decidability of $\mathfrak{m}$, find the first coefficient c_m that is a unit, and divide by it. The coefficients below m are in $\mathfrak{m}$, so this gives $t^m(1-tv) \in I$. Take $N = m+1$. From $t^N \equiv t^N(tv)$ we get $t^N \equiv t^N(tv)^N$, and multiplying by v^N gives $e \equiv e^2$. □

B is finite, so s satisfies a monic equation (determinant trick on $b_1, \ldots, b_r$). Lemma 3.1 with $I = \mathfrak{m}B$ and $t = s$ gives an idempotent e modulo $\mathfrak{m}B$. Put $y = xe$. Then $y \notin \mathfrak{q}$: $x \notin \mathfrak{q}$, and $e \notin \mathfrak{q}$ because $s^N(1-e) \in \mathfrak{q}$. For $b \in B$, by (2.1) and idempotency, modulo $\mathfrak{m}B$,

$$yb = xeb \equiv xe\,C(x) \equiv xe\,C(xe) = (XC)(y),$$

so $yB \subseteq R[y] + \mathfrak{m}B$. In coordinates, $yb_j = c_j + \sum_l \mu_{jl} b_l$ with $c_j \in R[y]$ and $\mu_{jl} \in \mathfrak{m}$.

Lemma 3.2 (Determinant lemma)**.** *Let $p = \det(y\delta_{jl} - \mu_{jl})$. Then $pb_j \in R[y]$ for all j, hence $pB \subseteq R[y]$. For every ideal $J \supseteq \mathfrak{m}B$, $p \equiv y^r \pmod{J}$; in particular $p \notin \mathfrak{q}$.*

Proof. Multiply the system $(yI - \mu)b = c$ by the adjugate. The congruence holds because $\mu \equiv 0$ modulo J, and $y \notin \mathfrak{q}$ with $\mathfrak{q}$ prime. □

Put $g = ps \notin \mathfrak{q}$ and choose $G \in R[X]$ with $G(y) = g$. Then $gB \subseteq R[y]$, and every $z \in E$ satisfies $g^k z = r(y)$ for some k and $r \in R[X]$: if $s^k z = b$, then $g^{k+1} z = p^k \cdot p(bs) \in R[y]$. We call this property *Full*.

4. The special fibre, read in E

Instead of working in E_g we work in E with the ideal

$$J = \{w \in E \mid g^k w \in \mathfrak{m}E \text{ for some } k\},$$

which is $\mathfrak{m}E_g \cap E$ read inside E. Then $J \subseteq \mathfrak{q}$. We take $H = P$ (the note uses $P/\gcd(P, X)$), and $W = G^n H$ for a suitable n.

Lemma 4.1. (1) *$H(y) \in J$;*
(2) *$H'(y)$ is a unit modulo J;*
(3) *g is a unit modulo J;*
(4) *$W(y) = \sum_j \mu_j R_j(y)$ with $\mu_j \in \mathfrak{m}$ and $R_j \in R[X]$.*

Proof. Since $s^N(1-e) \in \mathfrak{m}B$ and g is a multiple of s, we get $1 - e \in J$, so $y \equiv x$ modulo J. Hence (1) and (2) follow from (2.1); for (2), the congruence is carried through the dual numbers, where P' is the ε-component of $P(x + \varepsilon)$. For (3), g satisfies a monic equation, and Lemma 3.1 with $t = g$ gives $g^N(1 - gv) \in \mathfrak{m}B$. For (4), $g^k H(y) \in \mathfrak{m}E$. Write the elements of $\mathfrak{m}E$ involved as $\sum \mu_j z_j$, and clear their denominators by Full, one generator at a time. Both sides then lie in $R[y]$. □

5. The loop

Write $[c]_e = \sum_{i<e} c_i X^i$. A monic relation of degree e for y is $y^e = [a]_e(y)$, and $Q = X^e - [a]_e$. Let $V_e(\mu) = \sum_{i<e} \mu_i y^i$.

Proposition 5.1. *Let Q_0 be a monic relation for y and suppose $W(y) = \sum_j \mu_j R_j(y)$ with $\mu_j \in \mathfrak{m}$. There are a monic relation Q with $Q(y) = 0$, a polynomial U, and $c_0, \ldots, c_{e-1} \in \mathfrak{m}$ with*

$$W = UQ + [c]_e \quad \text{in } R[X].$$

Thus $\bar{Q}$ divides $\bar{W}$ in $k[X]$.

Proof. (a) *Reduction.* Every F is $UQ + [c]_e$ (Horner's scheme, using $X^e = Q + [a]_e$).

(b) *The $\mathfrak{m}$-part.* The set of $V_e(\mu)$ with all $\mu_i \in \mathfrak{m}$ is an $R[y]$-submodule; for multiplication by y, use $y^e = [a]_e(y)$. So $W(y)$ belongs to it.

(c) *One step.* Reduce $W = UQ + [c]_e$. Scan $c_{e-1}, \dots, c_0$ with the decidability of $\mathfrak{m}$. If all are in $\mathfrak{m}$, stop. Otherwise let $d < e$ be the top index with c_d a unit. Then $D = X^d + c_d^{-1}[c]_d$ is monic, and by (b), $D(y)$ lies in the $\mathfrak{m}$-part. Hence every y^j is $V_d(\beta_j)$ plus an element of the $\mathfrak{m}$-part. For $j < e$ this is a system $(1 - \mu)v = n$ with $v_j = y^j$ and $\mu \equiv 0$ modulo $\mathfrak{m}$. The determinant $\det(1 - \mu) \equiv 1$ is a unit because R is local, and the adjugate gives $y^j \in V_d(R)$. For $j = d$ this is a new monic relation of degree $d < e$; the same formulas move $W(y)$ into the new $\mathfrak{m}$-part.

The degree strictly decreases, so the loop terminates. □

The note takes $D_l = \gcd(\bar{Q}_l, \bar{G}^n \bar{H})$. Here D_l is the remainder of W modulo Q_l, made monic at its top unit coefficient; the loop thereby performs Euclid's algorithm itself.

6. The derivative

Lemma 6.1. *$U(y)Q'(y) \equiv G(y)^n H'(y) \pmod{J}$. Hence $Q'(y)$ is a unit modulo J, and $Q'(y) \notin \mathfrak{q}$.*

Proof. Evaluate $W = UQ + [c]_e$ at $y + \varepsilon$ in $E[\varepsilon]/(\varepsilon^2)$ and compare ε-components. We have $Q(y) = 0$, the derivative of $[c]_e$ has coefficients in $\mathfrak{m}$, and $W' = nG^{n-1}G'H + G^n H'$ with $H(y) \in J$. The right side is a unit modulo J by Lemma 4.1, and $J \subseteq \mathfrak{q}$. □

Neither $\bar{H} \mid \bar{Q}$ nor the factor $D = \bar{Q}/\bar{H}$ of the note is needed.

Put $g_1 = g\,Q'(y) \notin \mathfrak{q}$, $T = E[1/g_1]$, and

$$S = (R[X]/(Q))[1/(GQ')(a)].$$

The element y is a point of (Q, G) in T, so it defines $u : S \to T$ with $u(a) = y$.

7. The étale case

Lemma 7.1. (1) *u is surjective.*

(2) *Let $x_1, \dots, x_v$, $f_1, \dots, f_w$ be a presentation of E, with $x_i = \sigma_i(y)/g^k$ by Full and $y = Y(x)$. Then $\ker u$ is generated by $f_j(\sigma(a))$ and $a - Y(\sigma(a))$, where $\sigma_i(a)$ means $\sigma_i(a)/G(a)^k$.*

(3) *T has the lifting property.*

Proof. (1) By Full, $z \in E$ is $r(y)/g^k = u(r(a)G(a)^{-k})$, and $1/g_1 = u(1/(GQ')(a))$. (2) Let K be the displayed ideal. The map $E \to S/K$, $x_i \mapsto \sigma_i(a)$, kills the f_j and sends y to a and g_1 to a unit. So it extends to $\psi : T \to S/K$. Then ψu is the projection, since both send a to a. Hence $u(z) = 0$ implies $z \in K$. (3) A map $T \to C/N$ gives $E \to C/N$, which lifts to $E \to C$ because E is étale. The image of g_1 is a unit modulo the square-zero ideal N, hence a unit, so the lift extends to T. □

Proposition 7.2. *Let $I = \ker u$. Then $I = I^2$, and there is an idempotent f with $I = (1 - f)$. Consequently $T \cong S[1/f]$ is standard étale, for (Q, GF) where $f\,(GQ')(a)^n = F(a)$.*

Proof. Let $C = S/I^2$ and let N be the image of I, which is square-zero. Then $C/N = T$, and lifting gives $\sigma : T \to C$. The projection $S \to C$ and σu agree modulo N. Since S is unramified they are equal, so I maps to 0, that is, $I \subseteq I^2$. With generators $i_j = \sum_l a_{jl} i_l$, $a_{jl} \in I$, put $f = \det(1 - a)$. Then $fI = 0$ and $1 - f \in I$, so $f^2 = f$ and $I = (1 - f)$. Maps out of S that kill I are exactly those inverting f, equivalently $F(a)$. So T has the universal property of $S[1/f]$, which is standard for (Q, GF). □

Proof of Theorem 1.1. Take $s' = g_1 \notin \mathfrak{q}$. Then $E[1/s'] = T$ is standard étale by Proposition 7.2. □

8. The flat and unramified case

An R-module M is *flat* if every relation is explained [5]: if $\sum_i a_i m_i = 0$ with $a_i \in R$ and $m_i \in M$, there are $c_{ij} \in R$ and $n_j \in M$ with $m_i = \sum_j c_{ij} n_j$ and $\sum_i a_i c_{ij} = 0$ for all j.

Theorem 8.1. *Let E be a finitely presented, flat and unramified R-algebra and $\mathfrak{q}$ as in Theorem 1.1. There is $s' \in E \setminus \mathfrak{q}$ such that $E[1/s']$ is standard étale over R, hence étale.*

Sections 2–6 and Lemma 7.1(1),(2) do not use the lifting property. Unramifiedness enters only through the field case. They give $u : S \to T = E[1/g_1]$, surjective, with finitely generated kernel I. As a localisation of E, T is flat over R.

Proposition 8.2 (Transfer)**.** *Let S be standard étale for (Q, G), with point a, and let $\varphi : S \to T$ be a map of R-algebras with T flat over R. Then T is flat over S.*

Proof. Put $D = (T[X]/(Q))[1/(GQ')(X)]$, that is $S \otimes_R T$. It is flat over S, by flat base change through polynomials, a principal quotient and a localisation. There are maps $\lambda : S \to D$, $a \mapsto X$, $\rho : T \to D$ and $\mu : D \to T$, $X \mapsto \varphi(a)$, with $\mu\rho = \mathrm{id}$. Write $\alpha = \lambda(a)$ and $\beta = \rho(\varphi(a))$. Both are roots of Q in D, and $Q'(\alpha)$ is a unit. By synthetic division $Q = (X - \alpha)H$ in $D[X]$, so $H(\alpha) = Q'(\alpha)$ and $(\beta - \alpha)H(\beta) = 0$. Put $e = H(\beta)/H(\alpha)$. Then $e\alpha = e\beta$, and $\mu(e) = 1$ because μ sends α and β to $\varphi(a)$. In $D/\mathrm{Ann}(e)$ the images of α and β are equal. So the two maps $S \to D/\mathrm{Ann}(e)$ induced by λ and $\rho\varphi$ agree at a, hence are equal by the universal property of S. Thus $e\lambda(x) = e\rho(\varphi(x))$ for every $x \in S$. Therefore $t \mapsto e\rho(t)$ is S-linear and a section of μ. So T is an S-module retract of D, and flat. The element e need not be idempotent. □

Lemma 8.3 (Flat quotient)**.** *If $u : S \to T$ is surjective and T is flat over S, then every $x \in \ker u$ satisfies $x = xb$ for some $b \in \ker u$. Hence $I = I^2$.*

Proof. The relation $x \cdot 1 = 0$ in T is explained: $1 = \sum_j u(c_j)t_j$ with $xc_j = 0$. Choose t'_j with $u(t'_j) = t_j$, and put $b = 1 - \sum_j c_j t'_j$. Then $u(b) = 0$ and $xb = x$. □

Proof of Theorem 8.1. By Proposition 8.2 and Lemma 8.3, $I = I^2$. The end of the proof of Proposition 7.2 gives an idempotent f with $I = (1 - f)$, so $T \cong S[1/f]$ is standard étale. Take $s' = g_1$. □

A standard étale algebra is étale. Uniqueness of lifts is the unramifiedness of Section 1. For existence, let $a \in C$ reduce to a point modulo N, with $N^2 = 0$, and let w be an inverse of $Q'(a)$ modulo N. Then $b = a - Q(a)w$ satisfies $Q(b) = 0$ by Taylor's formula, since $Q(a) \in N$. Also $(GQ')(b)$ is a unit, since it is a unit modulo N.

The note proceeds differently. It injects $I/\mathfrak{m}I$ into $S/\mathfrak{m}S$ by flatness, uses the special-fibre isomorphism $S/\mathfrak{m}S \cong T/\mathfrak{m}T$ to get $I = \mathfrak{m}I$, and then applies Nakayama's lemma, which needs a further localisation. Here neither the special-fibre isomorphism nor a further localisation is needed. Flatness cannot be omitted: $\mathbb{Z} \to \mathbb{Z}/2\mathbb{Z}$ is surjective, finitely presented and unramified, and its kernel (2) is not idempotent.

9. Examples

Localisation. If $t \in \mathfrak{m}$, then $E = R[X]/(1 - Xt)$ is $(R[Y]/(Y))[1/t]$: points are x with $xt = 1$, and these correspond to t being a unit. Its special fibre is zero.

Adding a simple root. Let $P = aX^2 + bX + c$ and $E = (R[X]/(P))[1/P'(x)]$, with $D = P'(x) = 2ax + b$. Put $y = ax + b$, $s = yD$, $F = Y^2 - bY + ac$ and $G = Y(2Y - b)$. Then $x \mapsto -c/Y$ and $Y \mapsto y$ identify $E[1/s]$ with $(R[Y]/(F))[1/G]$, which is standard since $F' = 2Y - b$. Where a is a unit, P/a is monic, which gives a second chart. The charts cover E, by $1 = ax/D + s/D^2$. Inverting Y is necessary: for $R = \mathbb{Z}$ and $P = 2X^2 + X$, $Y = 0$ is a point of $(R[Y]/(F))[1/F']$ but not of $E[1/s]$.

10. The formalisation

10.1. **Where and how.** The three original local developments are checked by Unit (`https://github.com/coquand/unit`):

- `unit/sag/etale-local`: Theorem 1.1 with ZMT as a hypothesis, in the weak form of Section 2 (`TheoremLocal`);
- `unit/sag/etale-local-zmt`: this development and `unit/sag/zmt-general` (the proof of ZMT [3]) in one tree; the shared library files are the same files. The module `TheoremLocalZMT` discharges the ZMT hypothesis with `FinalZMTGeneral.Theorem`;
- `unit/sag/etale-flat-unramified`: Theorem 8.1, with the same weak ZMT and field case as hypotheses (`FlatTheoremLocal`).

The subsequent field and local-adapter developments are:

- `unit/sag/etale-field`: finite span, an actual finite basis and perfect trace from finite presentation and formal unramifiedness;
- `unit/sag/etale-field-complete`: the selected separable monogenic neighbourhood, including both large and small field branches;
- `unit/sag/etale-field-product`: the full finite product of separable monic quotients, without any prime parameter;
- `unit/sag/etale-field-local-interface`: the local proof with one selected factor and only finitely many generation identities;
- `unit/sag/etale-field-local-complete`: the completed field theorem applied to the exact localized special fibre, followed by the coefficient and denominator lifting bridge.

All these developments are parametrised by the segment `Theory` of `sag/`: univalence (`univ`) and replacement (`images`, `surjections`), with function extensionality and propositional univalence derived. No new axiom is introduced. On 30 October 2026 fresh checks gave: `etale-local`, 3985 declarations in 89 modules, 0 failed, 1.30 s; `etale-local-zmt`, 21057 declarations in 394 modules, 0 failed, 14.27 s. On 31 October 2026: `etale-flat-unramified`, 7076 declarations in 162 modules, 0 failed, 2.43 s. Each file starts with its informal proof.

The fresh check of `etale-field-complete` gives 17910 declarations, 0 failed, in 9.87 s; its 421 modules also pass `segc all`. All 68 named declarations in the final assembly have clean opening audits. The combined local adapter, `etale-field-local-complete`, gives 21221 declarations, 0 failed, in 11.72 s, with all 474 modules checked and all 86 named new declarations covered by the opening audit. These are complete import-tree counts; the owned declarations in the two final directories number 72 and 90, respectively. The full product development gives 18757 declarations, 0 failed, in 9.42 s, with 847 new declarations in 25 modules; all 446 modules pass `segc all`. Its opening audit covers all 797 named new declarations. The two complementary pieces are both retained in the small-field branch; the $\mathbb{F}_2^3$ regression exercises this construction.

10.2. **The original local statements.** The original ZMT-integrated theorem is

`TheoremLocalZMT.Over[...].FieldCase[field].theorem :  || Conclusion ||`

with `Conclusion` from `Local.Statement`:

$$(s' : E) \times \neg \mathfrak{q}(s') \times (T,\ i : E \to T) \times \texttt{IsLocalisation}\ T\ i\ s' \times (Q\ G) \times \texttt{Standard}_{Q,G}\ T.$$

`IsLocalisation` is the universal property of $E[1/s']$: $i(s')$ is a unit, and restriction along i is an equivalence from maps $T \to C$ to maps $E \to C$ inverting s'. `Standard` is the universal property of $(R[X]/(Q))[1/(GQ')(a)]$ by its points (Section 1). Thus isomorphisms of the note are stated by points, which identifies the algebras by Yoneda.

The parameters of `TheoremLocalZMT.Over` are:

- R with $\mathfrak{m}$ given by its closure properties, membership a proposition (`m-prop`), decidability `dec` and locality `rad`;
- E, the prime $\mathfrak{q}$ (closure, `q-prime`, `q-proper`, `q-m`: $\mathfrak{m}E \subseteq \mathfrak{q}$) with decidable membership `q-dec`;
- `etale`: E formally étale (`Lift`, `Unique`) together with a presentation; its generators are the x_i given to ZMT;
- `fo`: merely, $E/\mathfrak{m}E$ is a finite $R/\mathfrak{m}$-module (the hypothesis of ZMT).

The parameter of `FieldCase` is `field`: for every output c of ZMT (an s with $s-1 \in \mathfrak{m}E$, finiteness of $B = R[s, sx_i]$ and $B[1/s] = E[1/s]$), merely a `FieldData`: the ε_i, the identity `sum-one`, and for each factor a `Piece`: P with the three facts (2.1) multiplied by ε_i.

The intermediate theorem `TheoremLocal.Over[...].theorem-of` has, in place of `etale`'s generators and `fo`, the weak ZMT of Section 2: B, j, generators `bs`; merely the coordinates of each element (`coord`); s with merely $j(s) - 1 \in \mathfrak{m}E$ (`s-one`); and, for each z, merely k and b with $j(s)^k z = j(b)$ (`zmt`).

For Theorem 8.1 the statement is

```
FlatTheoremLocal.Over[...].theorem fd :  || Conclusion ||
```

with `fd` merely a `FieldData`. Its parameters are those of `TheoremLocal.Over`, with `etale` replaced by a presentation of E and `flat`: E is flat over R in the sense above. `Conclusion`, from `FlatLocalStatement`, is the conclusion above together with `Etale` T. The variant `from-unramified` takes `Unique` E and a principle `Unique` $E \to \|$`FieldData`$\|$, which is how the field case uses unramifiedness.

10.3. **The completed field theorem and its local adapter.** The selected field endpoint used by this local adapter is

```
UnramifiedFieldSelected.Over[k,setk,field]
      .At[C,fp,unique,prime].theorem.
```

Here k is a discrete nonzero field, C a k-algebra with merely a finite presentation, `unique` is uniqueness of square-zero lifting, and `prime` is a specified decidable proper prime $\mathfrak{p}$ of C. The theorem merely constructs a finite basis $b_1, \dots, b_n$ and a certificate

$$e^2 = e, \quad e \notin \mathfrak{p}, \quad eP(z) = 0, \quad e(P'(z)w - 1) = 0, \quad e(b_i - C_i(z)) = 0,$$

where P is monic and $C_i \in k[T]$. This is exactly `SelectedFactorStatement.Selected`. It gives generation on C_e and a separable monic annihilator. It does not assert that the displayed P presents C_e as $k[T]/(P)$, and no such assertion is needed. The basis, trace pairing, diagonal idempotent, monic equations and primitive element are constructed; none is a theorem parameter. There is no ZMT or assumed field-case principle in this theorem.

The proof uses the same diagonal certificate for flatness transfer, regularity and trace duality. For the primitive-element step, collect the coefficients of constructed monic annihilators of the basis vectors. Bounded coefficient closure either supplies sufficiently many distinct scalars or an actual finite coefficient subfield. In the first branch, polynomial and matrix McCoy, a determinant bound and the root bound construct z and w with

$$(z \otimes 1 - 1 \otimes z)w = 1 - \epsilon.$$

Tensor contraction gives generation after one selected idempotent refinement. In the second branch, the monic equations lift to the finite subfield; the finite image subalgebra, its selected field factor and a primitive element are constructed. This includes small fields, with no decision whether the whole field is finite. The full proof and its intermediate interfaces are recorded in [6].

For the local application the actual rings are

$$k = R/\mathfrak{m}, \qquad C = (B/\mathfrak{m}B)[1/s].$$

The checked endpoint is

```
LocalFieldComplete.Over[...,prop-m]
    .GivenInverse[N,v,relation]
.Field[field,fp,unique,prime,prime-map].theorem.
```

It applies the field theorem to this exact C, eliminates the resulting truncation into the selected-data proposition, and invokes the checked local proof. A basis or selected factor is no longer supplied. Its remaining explicit inputs include:

- the local setup, finite presentation and flatness of E, and the weak ZMT data $B \to E$, its finite span, s and denominator clearing;
- N, v with $s^N(1 - sv) \in \mathfrak{m}B$;
- the discrete field structure of this k, finite presentation and formal unramifiedness of this C;
- a decidable proper prime of C, compatible with $\mathfrak{q}$ in the direction $\mathfrak{q}(j(b)) \Rightarrow \mathfrak{p}(\eta(b))$ for $b \in B$.

In the coefficient bridge, each equality pulled back from C retains an explicit power of s. The map $B/\mathfrak{m}B \to C$ is never assumed injective. The regression takes $B = \mathbf{F}_2 \times \mathbf{F}_2$, $s = (1, 0)$ and $E = \mathbf{F}_2$; the nonzero class of $(0, 1)$ is killed by localization. It constructs the exact fibre's presentation and unramifiedness, then obtains its basis and selected factor from the new theorem. These results do not yet discharge all the listed local inputs from finite presentation, flatness and unramifiedness of E alone.

10.4. **Correspondence.**

Mathematics	Unit module
Definitions, Section 1	`FormallyEtale` (`SquareZero`, `Lift`, `Unique`, `Etale`), `StandardEtale` (`Points`, `Standard`, Taylor in dual numbers, `unramified`)
Choice of a factor	`TheoremLocal` (`find`, `Refine`, `FieldCase.At`)
x versus $x - 1$, $y = xe$, system	`Introduction` (steps 0–3), `HornerValues`
Lemma 3.1	`ResidualIdempotent` (over a discrete field), `IdempotentModM` (modulo an ideal $\supseteq \mathfrak{m}$)
Lemma 3.2	`DeterminantLemma` (`System`, `Congruence`, `Prime.p-out`), on `sag/exterior`
Full, Lemma 4.1	`Introduction` (step 4, `Jsat`), `ClearDenominators`, `IdealGenerated`, `LoopInputs`
Proposition 5.1	`PowerSums`, `MonicReduce` (a), `MPart` (b), `LoopStep`, `LoopNakayama` (c), `Loop`
Lemma 6.1	`DerivativeStep` (`Qprime`), `LoopQprime`
S	`StandardAlgebra` (built from a presentation and `Localization`)
Lemma 7.1	`Local`, steps (b)–(e)
Proposition 7.2	`EtaleCase` (`Surjection`, `Square`, `Localised`), `IdempotentIdeal`
Theorem 1.1	`Local.Field.theorem`, `TheoremLocal`
ZMT plugged in	`TheoremLocalZMT` (`At: coord`, `s-one`, `zmt`), on `FinalZMTGeneral.Theorem`, `CertificateSubalgebra`, `CertificateLocalisation.Iso.cleared`
Section 8: flatness	`FlatRelations` (explained relations), `FlatSystems`, `FlatPolynomialBaseChange`, `FlatPrincipalQuotient`, `FlatPowerLocalization`
Proposition 8.2	`FlatStandardChart` (D), `FlatRootFactor` (synthetic division), `FlatSeparableRetraction` (e), `FlatRetraction`, `FlatStandardTransfer`

Mathematics	Unit module
Lemma 8.3	`FlatQuotient`
Theorem 8.1	`FlatBaseStandardQuotient`, `FlatLocal`, `FlatTheoremLocal`
Standard implies étale	`StandardEtaleLift`
Selected field theorem	`UnramifiedFieldSelected`, `SelectedFactorComplete`
Finite selected-factor interface	`SelectedFlatLocal`
Coefficient and denominator bridge	`SelectedLocalLift`, `QuotientPowerClearing`
Field theorem applied to the exact fibre	`LocalFieldComplete`
Examples	`LocalisationExample`, `SimpleRootExample` (with `cover` and `Necessity`)

10.5. **Changes with respect to the note.** The proof above already incorporates these changes. We list them for a reader of `zmt2-revised.tex`.

(1) *Hypotheses used.* Of ZMT only $s \in 1 + \mathfrak{m}E$ and $s^k z \in B$ are used, and $B \to E$ need not be injective. Of the field case only the identity $\sum \varepsilon_i \equiv 1$ and the three facts of each factor are used. The factor and the refinement $s' = s\varepsilon_i$ are chosen in the original proof, using decidability of $\mathfrak{q}$. The newer field theorem produces one selected factor directly and the refined local interface retains only finitely many generation identities.
(2) *Lemma 1.* The hypothesis is any relation with a unit coefficient, and $N = m + 1$ always. The lemma is applied modulo $\mathfrak{m}B$ without forming $A = B/\mathfrak{m}B$.
(3) $H = P$. The gcd with X is not needed, because y is a unit modulo J.
(4) *Section 1 in E.* The note's $\mathfrak{m}E_g$ becomes the saturation J in E, so no localisation is constructed for Section 1.
(5) *Loop.* Remainders of W replace $\gcd(\bar{Q}_l, \bar{W})$.
(6) *Derivative.* $Q'(y)$ is a unit modulo J by the ε-component identity, without $\bar{H} \mid \bar{Q}$ and without the special-fibre isomorphism.
(7) *Kernel of u.* The note obtains finite generation from finite presentation. Here explicit generators are constructed from the presentation of E (Lemma 7.1(2)).
(8) *Mere existence.* Lists of coefficients for the polynomials of Lemma 3.2, expressions in generators, and certificates of $I = I^2$ are chosen under truncation. Hence the conclusion is truncated. I^2 must have propositional membership for the quotient ring.
(9) *Finitely many coordinates.* ZMT gives coordinates in B, the exponents k with $s^k z \in B$, and $s - 1 \in \mathfrak{m}E$ only merely. Full is therefore a mere property, and the proof chooses, in stages, the finitely many instances it uses: the coordinates of 1, s and sb_i; then (they depend on e) those of the elements $yb_i - (XC_i)(y)$ of $\mathfrak{m}B$, with coefficients in $\mathfrak{m}$; then (they depend on p) those of gb_i and yb_i; and Full at the generators of E, for the preimages in Lemma 7.1. Denominators are cleared (Lemma 4.1) by the same induction under the truncation.
(10) *The ZMT interface.* The B of ZMT is the image subalgebra $R[s, sx_i]$ of E. A zero generator is added to its module generators, so that there is at least one. The exponent k with $s^k z \in B$ is obtained by clearing the denominators of an expression of z in the x_i, not from the inverse of $B[1/s] \to E[1/s]$.
(11) *The flat case.* The special-fibre comparison and Nakayama's lemma are replaced by Proposition 8.2 and Lemma 8.3. The conclusion is reached at $s' = g_1$, without a further localisation.

10.6. **What remains.**

- *The remaining local connections.* The field theorem itself is now proved, including small fields. `FiniteSpecialFibre` also constructs the finite-fibre hypothesis from the actual residue quotient and transferred formal unramifiedness. What remains is to supply all the explicit inputs of `LocalFieldComplete` from the original local hypotheses: the actual localized fibre's presentation and unramifiedness, its compatible prime, the quasi-inverse relation, and the ZMT output. The old `FieldData` and `FieldCase` interfaces remain conditional. The full product decomposition is now proved in `etale-field-product`; supplying the local inputs and connecting that product to the original factor-selection argument remain tasks of local integration.
- *ZMT in the flat case* is still the weak hypothesis of Section 2; it has not been plugged in. Of `etale`, `TheoremLocalZMT` uses only the presentation of E, which the flat case also has.
- *Classical local–global step*: the separate segment `unit/sag/etale-classical-global` now turns primewise standard neighbourhoods over the original base into a finite cover, with the equation $\sum_i s_i a_i = 1$ and all chart certificates retained. Excluded middle and "every proper ideal lies in a prime ideal" are explicit parameters confined to that segment. The latter principle and excluded middle are derived from choice in the separate `unit/sag/choice-prime` development; `unit/sag/etale-choice-global` instantiates the segment with those derivations. The adapter accepts the exact local conclusion above; a stronger adapter retains monic polynomials explicitly, since the old conclusion does not record monicity. See `unit/doc/etale-classical-global.tex`.
- *Constructive finite-tree bridge*: the separate development `unit/sag/etale-dynamical-global` now extracts a finite certified cover from a radical/unit tree, retaining actual chart algebras without proposition resizing. It also constructs the first-unit scan used in the idempotent argument, proves standardness after inverting $F(x)$, and constructs the complementary standard charts at $F(x)$ and $1 - F(x)$. See `unit/doc/etale-dynamical-global.tex`.
- *Not formalised*: descent from the localised base (the lemma on changing the base from $R[1/f]$ to R and the finite-character lemma), their assembly with the local theorem and the finite cover, and the construction of the complete radical/unit tree from the original local hypotheses, in both the étale and the flat case; the remark on $R = k[t]_{(t)}$; and the identification $E \cong \mathbb{Z} \times \mathbb{Z}[1/2]$ in the simple-root example.

References

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[4] H. Lombardi, *Un théorème décisif*, arXiv:2506.07098, 2025.

[5] H. Lombardi and C. Quitté, *Commutative Algebra. Constructive Methods*, Springer, 2015.

[6] *The residue-field case: reducing the proof to what the local theorem uses*, `unit/doc/etale-field-reduction.tex`, with the checked field theorem in `unit/sag/etale-field-complete`.